Xiaoyu Fan

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Homepage

Educational background

Huazhong University of Science and Technology - School of AI & Automation - Intelligent Medical Engineering, Second Year of Undergraduate Studies | *Sep. 2024 - present*

Rank: 1 / 11 (Last Semester), 3 / 11 (Overall)

Core Courses: Data Structures and Algorithm Analysis, Databases, Computer Networks, **Machine Learning**, **Pattern Recognition**, **Computer Vision**, Medical Digital Signal Processing, Medical Image Processing, Anatomy and Physiology.

Research Experience

Shanghai Jiao Tong University, School of Artificial Intelligence, DeepDelta Lab - Prof. Shaobo Cui, RA | *May. 2026 - present*

Research Topic: **Causal World Model** – from perception to prediction, fault analysis and intervention.

What I focus on: Filling in the missing counterfactual pairs from human&robot data.

Union Hospital Affiliated to Tongji Medical College, HUST - Prof. Xiangdong Chen & Dong Yang's Group (Neuroscience-AI Interdisciplinary Direction), Scientific Collaboration | *Dec. 2025 - present*

In Feb 2026, I collaborated on the design of an algorithm for extracting features from **neural electrical signals**; Currently involved in the design of algorithms for **a patent relating to a quantitative pain prediction model in mice**, as well as the research and development of **a peripheral nerve electrical stimulation device based on an embedded AI chip**.

DeepkW Student Innovation Team, HUST, guided by Prof. Ran Wang | *Nov. 2025 - present*

Research **Multi-Agent Social Simulation** and **MoE**;
Joined the team for the **2025 SMP**;
Involved in **ACL 2026 Rebuttal**;

Skill

Language : C++, Python, Java, Matlab

Tools & Skills : Git, Vibe Coding, OpenCV, Transformer, Fine-tuning

Writing : Latex, Markdown, HTML

Honours and Awards

Huazhong University of Science and Technology

2024-2025 Self-Reliance and Endeavour Scholarship

Mastery of language

English (CET-6: 542)

Chinese (Mandarin, Level 2 Grade A)

Currently co-authoring a manuscript for **EMNLP 2026** submission on a collaborative writing framework, focusing on **personalised generation** and **cross-modal alignment**; Served as a primary reviewer for a paper at **IJCAI 2026.**

School of AI & Automation, HUST - Prof. Yang Xiao's Group (MLLM), Research Internship | *Jul. 2025 - present*

Conducted **MLLM intent understanding** research; Standardized eye-blinking behavior annotation for MLLM training during the summer holidays of 2025.

Project Experience

"Omission-based Deception"Fake News Detection Dataset | *Apr. 2026*

- Construction of a benchmark dataset (approximately 4,000 samples) for detecting "omission-based deception" in LLMs, featuring a hierarchical labelling system and six fine-grained categories; implementation of data generation scripts and evaluation workflows, supporting various modelling approaches such as SFT and prompting.

Image Segmentation and Region Coloring System (C++ / OpenCV) | *Mar. 2026 - Apr. 2026*

- Project Theme: Developed an end-to-end image processing pipeline, covering seed generation, watershed segmentation, and region graph coloring.
- Core work:
  • Designed a constraint-based sampling method to generate seed points with controlled minimum distance
  • Constructed a Region Adjacency Graph (RAG) to model relationships between regions
- Achievements:
  • Achieved second-level runtime on 600×600 images with ~1000 seeds
  • Maintained high coloring success rate

2026 MCM (Leader & Programmer), M Prize | *Feb. 2026*

- Project Theme: Dynamic Simulation and Prediction of Mobile Phone Lithium-Ion Battery Power Consumption Driven by Multiple Factors.
- Core work: Developed a continuous-time battery model with bidirectional thermo-electrical coupling based on the **Arrhenius Equation**; Implemented numerical solvers and utilized **Markov Chains** to quantify user behavior uncertainty, achieving dynamic SOC prediction.

2025 CUMCM (Leader & Programmer) | *Sep. 2025*

- Project Theme: Optimisation of NIPT Testing Timing Based on **Clinical Data** and Intelligent Assessment Model for Foetal Chromosomal Abnormalities.
- Core work: Data cleaning using Python (Pandas, Scikit-learn), constructing **LightGBM** models, and coordinating the team to

complete the entire workflow from modelling and programming to paper writing.

C Language Course Design — Resume Design Assistance System (Team Leader) | *Jan. 2025 - Apr. 2025*

- Project Theme: A Modular CV Design Assistance System Based on C Language and the Borland C++ 3.1 environment.
- Core work:Architected a **GUI engine from scratch** using BGI library; implemented 16/24-dot matrix Chinese rendering and structure-based data/display decoupling.
- Achievements: Implemented multi-format export functionality (plain text, simulated .doc), alongside integrated user registration and login systems, and a file-based local database management module.